

Learning from data

Machine learning

Machine learning is a branch of artificial intelligence in which computers learn patterns from data to make predictions or decisions.

Supervised learning

Supervised learning uses labelled training examples. Each example includes input features and a known target. The model learns a relationship between the inputs and targets.

Classification and regression

Classification predicts a category, such as spam or not spam. Regression predicts a number, such as a house price. A house price prediction model is a regression example.

Training and generalisation

Training, validation, and test sets

The training set is used to fit a model. The validation set helps select settings and compare models during development. A held-out test set evaluates the final model on data not used for fitting or selection.

Overfitting

Overfitting happens when a model learns the training data too closely, including noise, and performs poorly on new data. A large gap between good training performance and worse validation performance can indicate overfitting.

Reducing overfitting

Possible ways to reduce overfitting include using more representative data, regularisation, early stopping, and a simpler model. Keep the test set separate while choosing these settings.

Evaluating predictions

Accuracy

Accuracy is the number of correct predictions divided by the total number of predictions. It can be misleading when classes are highly imbalanced.

Precision

Precision is true positives divided by true positives plus false positives. For a spam classifier, it measures how many messages predicted as spam actually are spam.

Recall and F1

Recall is true positives divided by true positives plus false negatives. It measures how many actual positive examples the model finds. F1 is the harmonic mean of precision and recall.